VI. Numerical problems.

1. The frequency of a source of sound is 600 Hz. Calculate the numbner of times it vibrates in a minute?
2. A stone is dropped from the top of a tower 750 m high into a pond of water at the base of the tower. Calculate the number of seconds for the splash to be heard?
   (Given g = 10 m s^{-2} and speed of sound = 340 m s^{-1})

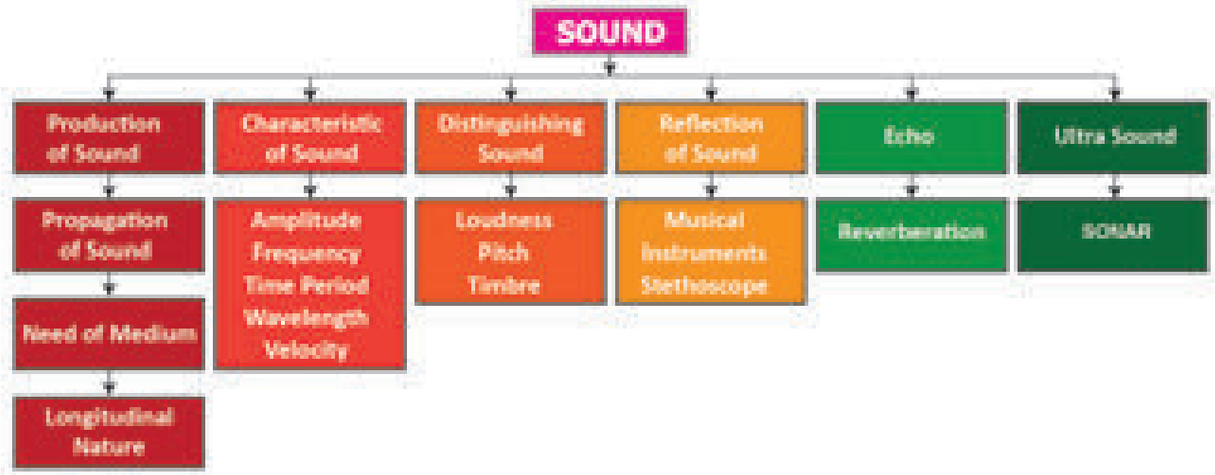
REFERENCE BOOKS

1. Applied Physics – Rajasekaran S and others – Vikas Publishing House Pvt Ltd.
2. Fundamentals of Physics – Halliday, Resnick and Walker, Wiley India Pvt Ltd. NewDelhi.

INTERNET RESOURCES

www.britannica.com/science/ultrasonics
https://www.searchencrypt.com
https://www.soundwaves.com/

Concept Map



ICT CORNER

Sound

Steps

- Type the given URL to reach "pHET Simulation" page and download the "java" file of Sound.
- Open the "java" file and plug in your headphone. Click "Audio enabled" box from right side to hear the sound waves.
- Switch the tabs from the top to simulate various properties of sound waves. Watch the longitudinal sound waves from different interfaces by altering "Frequency" and "Amplitude".
- Alter "Air Density" and observe its effect on the sound waves. Use "Reset" to repeat the experiment.

Sound Simulator

URL: https://phet.colorado.edu/en/simulation/legacy/sound or Scan the QR Code.

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